Multi-Horizon Time Series Forecasting of Non-Parametric CDFs with Deep Lattice Networks

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Time Series Forecasting

Using a past time series, a function is trained to predict a future time series.

$$\chi = (x_{t-w}, \dots, x_t), \text{ with } \chi \in \mathbb{R}^{w \times F}$$

$$\gamma = (y_{t+1}, \dots, y_{t+h}), \text{ with } \gamma \in \mathbb{R}^{h \times 1}$$

$$f(\chi) = \gamma$$

t	Time point
h	Number of horizons
F	Number of features
w	Length of window

Probabilistic Forecasting

Instead of predicting a single output series of timesteps, one can also predict a probability distribution over each step in the output series.

$$f(\chi) = P(\gamma)$$

Quantile Regression

QR is a non-parametric way of modeling a probability function across a predefined set of quantiles.

$$l_\tau(\gamma,\gamma')=\begin{cases}\tau(\gamma-\gamma') & \text{if } \gamma-\gamma'\geq 0,\\ (1-\tau)(\gamma'-\gamma) & \text{else.}\end{cases}$$

Simultaneous Quantile Regression

In SQR, the quantile is not predefined, but part of the input, and randomly sampled during training. When training long enough, we trained for all possible quantiles, yielding an approximation of a cumulative density function (CDF).

$$\tau \sim U(0,1) \qquad f(\chi, \tau) = \gamma_\tau$$

Quantile Cross-over

A neural network that is only trained to find quantiles as categorical output (QR), or single output, has no obligation to adhere to the correct ordering of quantiles.

$$f(\chi, \tau_a) \leq f(\chi, \tau_b) \text{ for } \tau_a \leq \tau_b.$$

For a legitimate CDF, we need a non-crossing quantile function.

Quantile Cross-over

Can be fixed.

- E.g. sum positive weights in output.

However, there are problems.

- Not for SQR, but only QR.
- No flexible output size.

What about monotone networks?

Deep Lattice Networks (DLNs)

Can be constrained monotonically in respect to an input for every layer in the network.

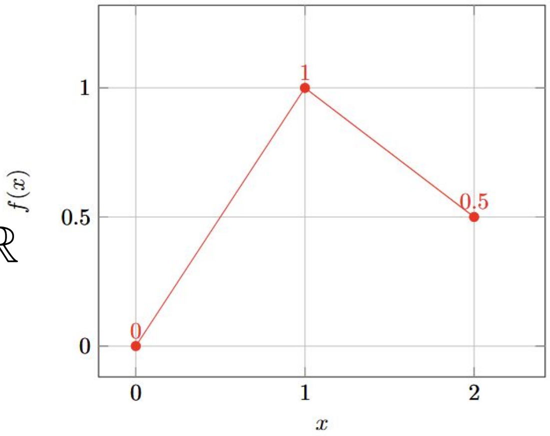
- Lattices are hypercubes of interpolated piecewise linear functions.
- A 1D lattice is an interpolated piecewise linear function.

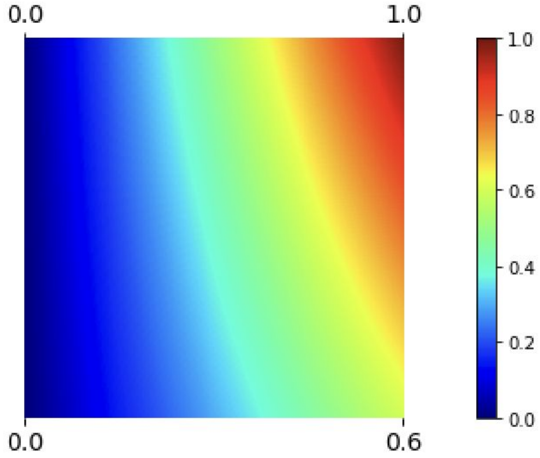
Lattice: 1- and 2-Dimensional

$$F : \mathbb{R}^D \to \mathbb{R}$$

$$f_{pwl}(\cdot\,; w) : [0, V-1] \to \mathbb{R}$$

, with V-1 vertices.





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https://openreview.net/forum?id=0pxiMpCyBtr

Lattices

a D-dimensional lattice of size $V \in \mathbb{N}^D$ consists of a regular *D*-dimensional grid of look-up table vertices

$$M_V = \{0, 1, ..., V[1] - 1\} \times ... \times \{0, 1, ..., V[D] - 1\}$$

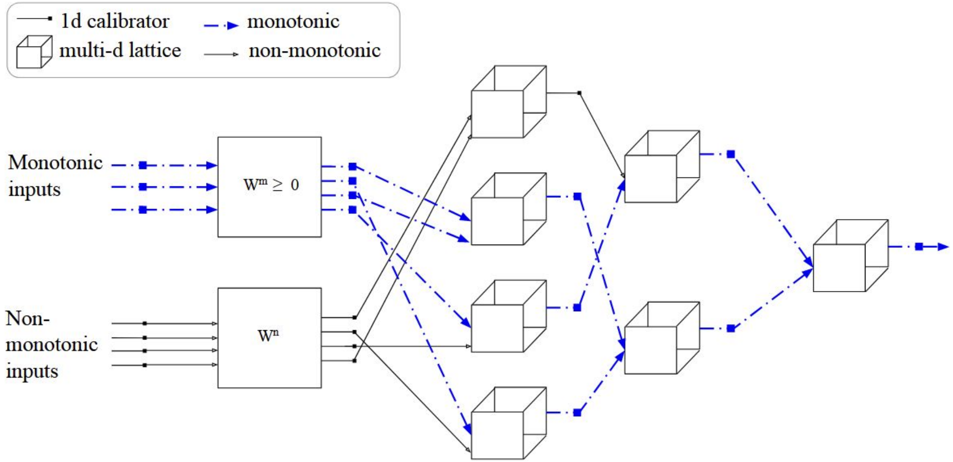
Given $f : D_v \to \mathbb{R}$, the resulting lattice model is defined by

$$f(x;\theta) = \theta \cdot \Phi(x) = \sum_{v \in M_V} \theta v(\Phi(x))_v$$

$$(\Phi_{multilinear} V(x))_v = \prod_{j \in [D]} f_{pwl}(x[j]; e_{v[j]+1, V[j]}), v \in M_V$$

DLNs

- Lattices are trainable multi-dim linear piecewise functions.
- Lattices model the complete input output relation function in linear pieces.
    - Explainable!
    - Constrainable!
    - But really big. v^D



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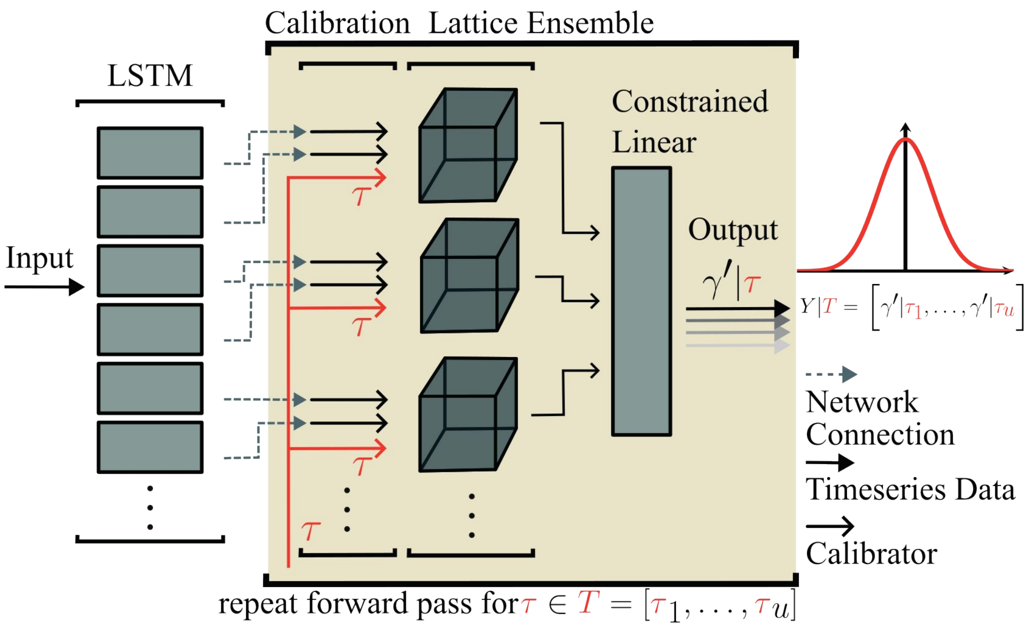
DLNs with Probabilistic Time Series

1. Adding an encoder for the timeseries. We used LSTM.
2. Feed into an ensemble of small lattices.
3. Feed sampled quantile values into all sublattices of the probabilistic decoder.
    a. During training, randomly sample quantiles for every minibatch.
    b. During testing, draw a specified set of quantiles which can portray the whole CDF.
4. Use a constrained linear layer to arrive at the necessary horizon length.



Solar Irradiance Forecasts

Ground Observations of 5 years of hourly data in southern Norway.
Input includes also past weather variables.
3 years for training
1 year for validation
1 year for testing

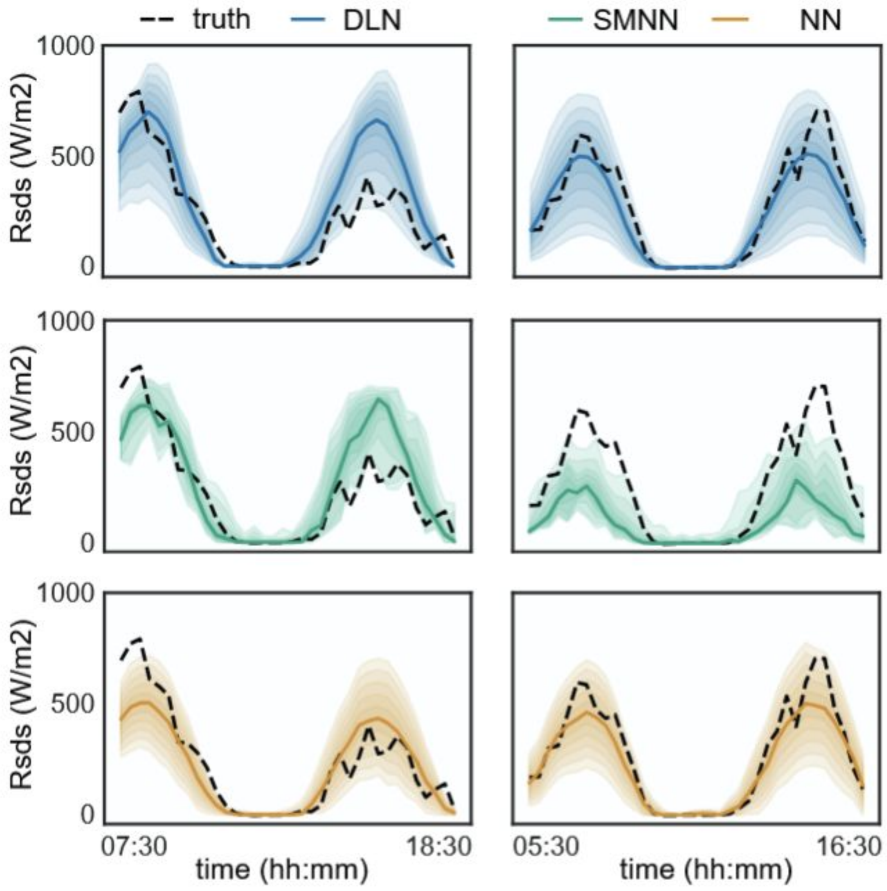
Horizon = 36 hours
Window = 96 hours

We forecast the irradiance at 1 station with past weather observations and past irradiance from 19 others.

Model comparisons

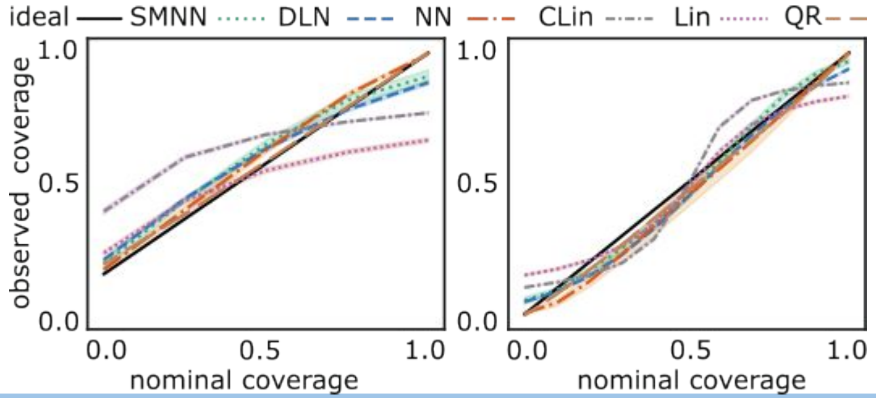
A smart persistence model as baseline.
Otherwise; LSTM + a decoder using SQR:
Linear layer (Lin)
Constrained linear layer (CLin)
Unconstrained feedforward neural network (NN)
Scalable monotonic neural network (SMNN)
Deep lattice Network (DLN)

Lastly, a decoder with NN and simple QR with a predefined set of quantiles.



36h forecasts for 2 different days of the test set.

PICP (left) and reliability diagram (right). Averaged across horizons.



Conclusions

DLN shows good performance against the unconstrained alternative, even though it is constrained.

DLN shows better performance than SMNN, a SoA monotonic network.

DLN has a couple other nice properties that could be exploited in future research:

- Not bound to only monotonicity
- More explainable
- But very large.

Thank you for your attention!

For questions, I'll leave my email here: niklase@uio.no

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